DiffWAM: Supplementary Material

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https://zgca-hmi-lab.github.io/DiffWAM/
https://github.com/ZGCA-HMI-Lab/DiffWAM

A Additional Experimental Details

The key configurations for the three experimental settings are summarized in Table 1. We use DiffWAM-S for the main LIBERO and real-world experiments and DiffWAM-L for the main RoboTwin 2.0 experiments.

Setting	LIBERO	RoboTwin 2.0	Real
Model configuration	DiffWAM-S	DiffWAM-L	DiffWAM-S
Training hardware	8× A100	8× A100	8× A100
Grad. accumulation	2	2	2
Weight decay	1×10^{-2}	1×10^{-2}	1×10^{-2}
LR schedule	Cosine	Cosine	Cosine
Learning rate	1×10^{-4}	1×10^{-5}	1×10^{-4}
Batch size	8	8	8
Training steps	1.3×10^{4}	3×10^{4}	6.3×10^{4}
Network depth N	30	30	30
Network depth M	10	10	10
Cameras	2	3	2
Video size	$[224, 448]$	$[384, 320]$	$[224, 576]$
Image resize	$[224, 224]$	$[240, 320]$	$[240, 320]$
Action dim.	7	14	7
Proprio dim.	8	14	7
Normalization	min/max	z-score	z-score
Validation split	0.0	0.01	0.0

Table 1. Implementation configurations for each experimental setting.

LIBERO

LIBERO [1] consists of four suites: LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, and LIBERO-Long. Each suite contains 10 tasks and provides 500 expert demonstrations, resulting in 40 tasks and 2,000 demonstrations in total. We conduct 100 rollouts per task, amounting to 4,000 evaluation trials, and report the average success rate for each suite and across all four suites. For this benchmark, DiffWAM-S is initialized from Wan2.2-5B and trained for 1.3×10^{4} steps with a learning rate of 1×10^{-4} and a batch size of 8. The model uses two camera views, a 7-dimensional action space, an 8-dimensional proprioceptive input, min/max normalization, and network depths of $N = 30$ and $M = 10$.

Method	Embodied PT	Clean	Random	Avg.
With Embodied Pretraining				
π_0 [3]	Yes	46.4	16.3	31.35
X-VLA [4]	Yes	68.0	20.9	44.45
$\pi_{0.5}$ [5]	Yes	70.7	46.0	58.35
Without Embodied Pretraining				
StarVLA [6]	No	46.5	3.2	24.85
AHA-WAM [7]	No	64.3	3.2	33.75
FastWAM† [8]	No	**70.2**	1.50	35.85
DiffWAM (Ours)	No	69.7	**5.27**	**37.47**

Table 2. **Clean-to-random generalization on RoboTwin 2.0.** Success rates (%) for policies trained on clean demonstrations and evaluated in clean and randomized environments without further adaptation. Embodied PT denotes additional embodied pretraining. External baseline values and their pretraining groups follow the comparison reported in PINT [9]; † denotes our own FastWAM evaluation, rather than the FastWAM reproduction reported there. Our clean-only runs do not start from a task-trained FastWAM checkpoint. Avg. is the mean of Clean and Random. For DiffWAM, each condition comprises 30 trials per task; Clean, Random, and Avg. are computed from the unrounded success rates before display rounding. Bold indicates the best value within the displayed non-pretrained group.

Method	Additional training	Clean success (%)
FastWAM (continued training)	1 epoch	88.47
DiffWAM (Ours)	1 epoch	**93.13**

Table 3. **Controlling for additional training on RoboTwin 2.0.** Both methods start from the same task-trained FastWAM checkpoint and receive one additional epoch of training on clean demonstrations under matched conditions. This experiment is separate from the clean-to-random study in Table 2.

RoboTwin 2.0

RoboTwin 2.0 [2] evaluates bimanual manipulation under both clean and heavily randomized environments. We adopt its multi-task setting with 50 tasks and use approximately 27k demonstration trajectories in total, comprising 2,500 demonstrations from clean scenes and 25,000 demonstrations with strong scene randomization. Each task is evaluated over 100 trials under both clean and randomized conditions, and we report the success rates for the two settings and their average. Due to computational resource constraints, DiffWAM-L is initialized from FastWAM for this benchmark and trained for 3×10^4 steps with a learning rate of 1×10^{-5} and a batch size of 8. The model uses three camera views, a 14-dimensional action space, a 14-dimensional proprioceptive input, z-score normalization, and network depths of $N = 30$ and $M = 10$.

Clean-to-random evaluation

The additional experiment in Table 2 uses clean demonstrations only during task training and evaluates the resulting policies under randomized RoboTwin 2.0 conditions without adaptation. These runs start task training without loading a task-trained FastWAM checkpoint; training from scratch here refers to robot-task training and does not imply that all model parameters are randomly initialized. This experiment is distinct from the full-data training configuration above, which includes randomized demonstrations. Clean/Random success is 70.2%/1.50% for our FastWAM baseline and 69.7%/5.27% for DiffWAM. We evaluate DiffWAM over 30 trials per task in each condition, totaling 1,500 trials per condition. It succeeds in 1,045 clean

trials and 79 randomized trials, yielding unrounded success rates of 1,045/1,500 and 79/1,500. The Clean/Random average is computed before rounding as 1,124/3,000, or 37.47% when rounded to two decimal places. The corresponding FastWAM average is 35.85%. Clean results from the separate checkpoint-continuation experiment are not paired with these values.

Checkpoint-continuation control

For Table 3, FastWAM and DiffWAM instead share the same task-trained FastWAM checkpoint as their starting point, with the additional Diff pathway introduced for DiffWAM. Both receive one additional epoch of training on clean demonstrations under matched conditions and are evaluated in the clean setting. The reported success rates are 88.47% and 93.13%, respectively. This comparison controls for the additional training budget and is separate from both the clean-to-random evaluation and the full-data RoboTwin result. No mechanistic claim about overfitting is made from the FastWAM score alone.

Real-world

We evaluate DiffWAM-S on the ALOHA-AgileX platform,[1] using approximately 1.1k demonstration trajectories in total across Place Pens, Store Block, and Stack Cups and reporting the execution success rate over 20 trials per task for each method (60 trials per method in total). For these experiments, DiffWAM-S is initialized from Wan2.2-5B and trained for 6.3×10^4 steps with a learning rate of 1×10^{-4} and a batch size of 8. The model uses two camera views, a 7-dimensional action space, a 7-dimensional proprioceptive input, z-score normalization, and network depths of $N = 30$ and $M = 10$.

B Additional Experimental Results

This section describes the comparison methods and provides task-level RoboTwin 2.0 results, real-world analyses, visualizations of state and visual difference prediction, and supplementary training-performance results.

B.1 Comparison Methods

We compare DiffWAM with representative vision language action (VLA) and world action model (WAM) approaches. VLA methods directly map visual-language observations to actions, while WAM methods additionally model action-conditioned dynamics.

VLAs

OpenVLA [10] and OpenVLA-O [11] are an open VLA baseline and its optimized variant; MemoryVLA [12] augments VLA control with memory. $\pi 0$ [3] and $\pi 0.5$ [5] are flow-based VLA models, with the latter targeting open-world generalization. GR00T-N1 [13] is a generalist humanoid foundation model, X-VLA [4] targets cross-embodiment transfer, ABot-M0 [14] learns an action manifold, and QwenVLA [15] unifies VLA modeling across tasks and embodiments.

WAMs

Motus [16] models latent actions in a unified world model; LingBot-VA [17] uses causal world modeling; and FastWAM [8] is the WAM baseline used for RoboTwin 2.0 initialization. ImageWAM [18], X-WAM [19], and WLA [20] respectively explore image editing based WAM, 4D world action modeling, and joint world language action modeling.

[1] https://github.com/agilexrobotics/aloha-agilex

B.2 RoboTwin 2.0 Results

Full data training

Table 6 reports task-level results when training with the full RoboTwin 2.0 data. DiffWAM-L achieves 93.28% success in clean scenes and 92.02% under randomization, for an average of 92.65%. This improves over the reproduced FastWAM baseline by 1.55 percentage points overall, with gains in both clean and randomized conditions (91.88% to 93.28% and 90.32% to 92.02%, respectively). At the task level, DiffWAM-L is competitive across a broad range of manipulation skills and reaches or ties the best result on many tasks, indicating that explicitly modeling inter-frame visual differences improves performance without sacrificing robustness to scene variation.

B.3 Independent Training Runs and Uncertainty

For each of LIBERO and the full-data and clean-only RoboTwin 2.0 settings, we independently train our method three times with different random seeds and evaluate each resulting model using the corresponding benchmark protocol. Table 4 reports the score from each training run, together with the mean and sample standard deviation across the three runs. The resulting scores are $92.52 \pm 0.11\%$ on full-data RoboTwin 2.0, $98.67 \pm 0.12\%$ on LIBERO, and $37.39 \pm 0.08\%$ for the Clean/Random average in the clean-only RoboTwin setting.

Benchmark	Metric	Run 1	Run 2	Run 3	Mean $\pm$ Std.
RoboTwin 2.0	Avg. (full-data)	92.49	92.65	92.43	92.52 ± 0.11
LIBERO	Avg.	98.60	98.80	98.60	98.67 ± 0.12
RoboTwin 2.0	Avg. (clean-only)	37.32	37.38	37.47	37.39 ± 0.08

Table 4. **Independent training runs of DiffWAM.** Success rates (%) from three models trained independently with different random seeds and evaluated separately. Run 1–3 denote independent training-and-evaluation runs. We report their arithmetic mean and sample standard deviation ($n = 3$, denominator $n - 1$). RoboTwin Avg. is the mean of Clean and Random success; full-data and clean-only denote the respective training settings. The clean-only average is not the Random-only OOD success rate. LIBERO Avg. is the average over its four suites.

The reported metrics exhibit limited variability across the three independently trained models, supporting the consistency of the method across training seeds in these settings. Each observation corresponds to a separate training run followed by evaluation, so the standard deviations summarize variation across complete training-and-evaluation runs. Main-table scores correspond to individual runs and are reported separately from the three-run means. In particular, the clean-only RoboTwin metric averages Clean and Random success and should not be interpreted as a Random-only success rate.

B.4 Real-world Results

Our real-world study evaluates DiffWAM-S on three manipulation tasks: Place Pens, Store Block, and Stack Cups. We use approximately 1.1k demonstration trajectories in total and train for 63k optimization steps. Each method is evaluated over 20 trials per task, totaling 60 trials across the three tasks. As shown in Figure 1 of the main paper, DiffWAM-S achieves success rates of 90%, 85%, and 90%, respectively, compared with 80%, 70%, and 70% for FastWAM. These rates correspond to 18/20, 17/20, and 18/20 successful trials for DiffWAM-S, compared with 16/20, 14/20, and 14/20 for FastWAM. The average success rate increases from 73.3% (44/60) to 88.3% (53/60), an absolute gain of 15.0 percentage points. The improvements across all three tasks support the effectiveness of joint state and visual difference modeling in this real-world setting. Broader evaluation across objects, environments, and embodiments remains necessary to establish generality.

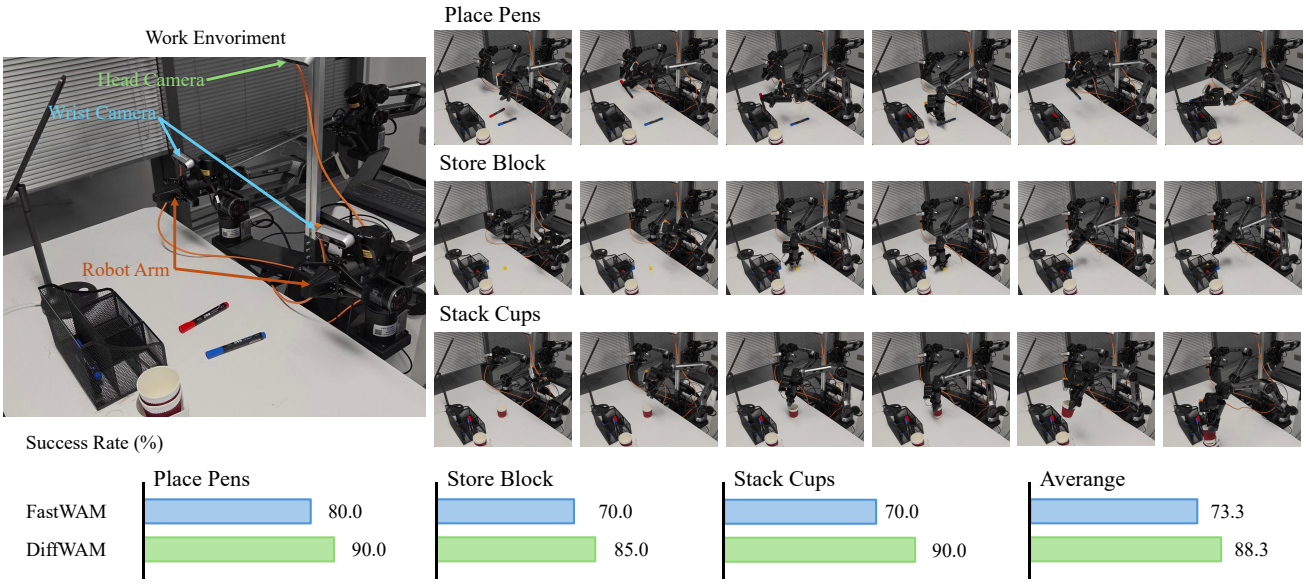


Figure 1. Real-world setup, task execution sequences, and success rates. Left: the robot workspace and camera placement. Right: example executions of Place Pens, Store Block, and Stack Cups. Bottom: success rates of FastWAM and DiffWAM-S across the three tasks. We use approximately 1.1k demonstration trajectories in total and train DiffWAM-S for 63k steps. Each method is evaluated over 20 trials per task, totaling 60 trials across the three tasks.

Metric	f_1	f_2	Δf_1	Δf_2
PSNR⇑	37.94	30.79	-	-
SSIM⇑	0.9851	0.9701	-	-
Latent Error⇓	-	-	0.0055	0.0136

Table 5. Future-state and visual difference prediction. PSNR and SSIM evaluate predicted future observations; latent error evaluates predicted differences between consecutive frame latents.

B.5 Visualization Results

Task Level Output Visualization

Figure 3 qualitatively visualizes the outputs of the World and Diff Experts on representative manipulation tasks. Both experts receive the same noised future-frame latents: the World Expert predicts future states through denoising, while the Diff Expert directly regresses differences between consecutive ground-truth frame latents. The predicted future observations preserve scene structure and reflect the progression of the manipulation trajectory. The visualized diff outputs highlight localized variation around the robot and manipulated objects, illustrating the complementary information captured by state and visual difference prediction. These differences describe inter-frame visual variation and do not exclusively isolate changes caused by robot actions.

Table 5 quantitatively evaluates the two prediction branches. For the first and second future observations, the World Expert achieves PSNR values of 37.94 and 30.79 and SSIM values of 0.9851 and 0.9701, respectively. The Diff Expert obtains latent prediction errors of 0.0055 and 0.0136 for Δf_1 and Δf_2. These results characterize the model's ability to predict both future



Figure 2. Supplementary per-suite training improvements of DiffWAM over the FastWAM baseline on LIBERO at each evaluation checkpoint.

scene states and their inter-frame latent differences.

B.6 Training Performance Analysis

Figure 2 reports the per-suite improvement of DiffWAM over the FastWAM baseline throughout training. DiffWAM achieves non-negative gains at all earlier evaluation checkpoints and improves the final average success rate. The advantage is strongest in the early and middle stages: the average gain reaches 14.9 points at 4k steps and 12.8 points at 6k steps. At 4k steps, the improvements on LIBERO-Object and LIBERO-Goal are 28.2 and 18.2 points, respectively; LIBERO-Long reaches its largest gain of 37.8 points at 6k steps. Although the gaps narrow as training proceeds, DiffWAM improves the final average success rate by 1.75 percentage points, with gains on Goal and Long, a tie on Object, and a 0.2-point decrease on Spatial, including a 6.0-point gain on LIBERO-Long. These results show that DiffWAM provides consistent training improvements, especially for the more challenging goal-conditioned and long-horizon tasks.

C Limitations and Future Work

C.1 Limitations

Several limitations remain. First, the independent M-layer Diff Expert pathway introduces additional computation and memory during training. Freezing the LDB in DiffWAM-S does not eliminate this cost, since diff predictions are still computed and their gradients propagate into the shared representation. Both the World and Diff Experts are removed at inference, so these visual prediction pathways add no online action-generation overhead. Second, the difference target captures inter-frame latent variation without explicitly distinguishing changes caused by robot actions from camera motion, illumination changes, or movement of unrelated objects. This ambiguity may be more significant in dynamic, unstructured environments. Third, the clean-to-random result establishes a benefit under the evaluated RoboTwin perturbations, while absolute Random success remains low and generalization to other shifts is not established. The experiments cover a limited range of data scales and embodiments, and DiffWAM is not pretrained on a large-scale robot dataset. Broader evaluation is needed to establish how the benefits of DiffWAM-S and DiffWAM-L generalize across training-data regimes and deployment settings.

C.2 Future Work

Future work will extend DiffWAM in three corresponding directions. First, more efficient Diff Expert designs, such as lightweight prediction heads or selective temporal token processing, could reduce the additional training cost and enable scaling to larger backbones, longer horizons, and richer multi-view observations. Second, we will investigate action-conditioned, motion-aware, and object-centric constraints to disentangle action-controllable changes from nuisance dynamics; multi-scale and multi-horizon diff prediction may further capture both fine-grained contact dynamics and longer-term task progress. Third, large-scale pretraining on diverse robot interaction data is a promising direction for learning more transferable visual and action priors before downstream adaptation. Evaluating this setting across different embodiments, camera configurations, objects, and long-horizon real-world tasks will be important for establishing the generality of diff prediction.

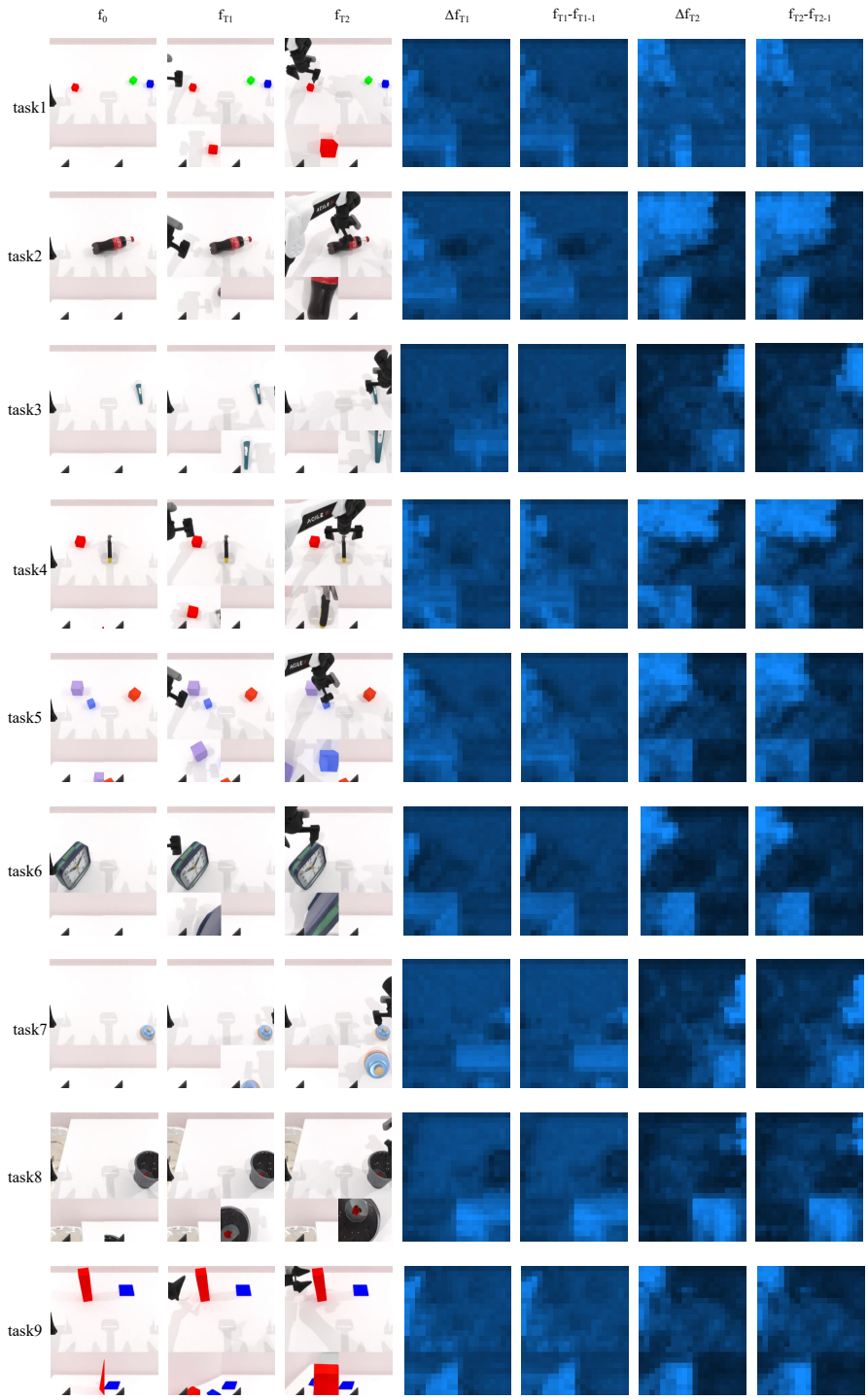


Figure 3. Qualitative visualization of future-state and visual difference prediction by the World and Diff Experts on representative manipulation tasks.

Task	π0.5		Motus		LingBot-VA		GigaWorld-Policy		X-WAM		FastWAM		DiffWAM-L	
	Clean	Random	Clean	Random	Clean	Random	Clean	Random	Clean	Random	Clean	Random	Clean	Random
Adjust Bottle	**100**	99	89	93	90	94	**100**	**100**	**100**	99	**100**	**100**	**100**	**100**
Beat Block Hammer	96	93	95	88	96	98	86	86	98	96	**99**	90	97	**100**
Blocks Ranking RGB	92	85	99	97	99	98	92	96	99	95	**100**	**100**	**100**	**100**
Blocks Ranking Size	49	26	75	63	94	96	44	48	76	82	94	**100**	**100**	97
Click Alarmclock	98	89	**100**	**100**	99	**100**	**100**	**100**	98	99	**100**	**100**	**100**	**100**
Click Bell	99	66	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
Dump Bin Bigbin	92	97	95	91	89	96	92	**100**	90	96	97	97	**100**	**100**
Grab Roller	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
Handover Block	66	57	86	73	**99**	78	80	80	88	79	95	83	97	**90**
Handover Mic	98	97	78	63	94	96	72	72	88	89	99	**100**	**100**	**100**
Lift Pot	18	17	38	38	40	28	16	12	**100**	99	58	97	**100**	**100**
Move Can Pot	96	85	96	**99**	**100**	**99**	98	98	80	84	**100**	80	90	87
Move Playingcard Away	51	55	34	74	94	97	76	78	**100**	98	90	**100**	**100**	**100**
Move Stapler Pad	84	61	93	96	99	**99**	90	90	67	70	**100**	53	73	57
Hanging Mug	96	84	**100**	96	**100**	**99**	78	72	46	55	**100**	53	63	57
Open Laptop	56	42	83	85	91	79	92	82	96	97	77	**100**	**100**	**100**
Open Microwave	90	96	95	91	92	94	96	**98**	89	92	**98**	37	73	53
Pick Diverse Bottles	34	77	**95**	91	82	86	74	66	91	**92**	62	87	93	87
Pick Dual Bottles	81	71	90	91	89	82	82	70	99	**100**	80	97	**100**	97
Place A2B Left	93	63	96	90	**100**	**99**	86	86	90	87	**100**	93	93	97
Place A2B Right	87	82	88	79	**97**	93	94	88	92	89	95	97	90	**100**
Place Bread Basket	87	84	91	87	**97**	**95**	90	92	90	91	93	87	87	90
Place Bread Skillet	77	64	91	94	**97**	95	82	82	90	96	91	**97**	**97**	93
Place Can Basket	85	66	86	83	**95**	**90**	94	**90**	84	82	90	60	63	77
Place Cans Plasticbox	94	87	98	98	97	95	98	96	99	98	71	**100**	**100**	**100**
Place Container Plate	62	62	81	76	81	84	78	74	98	**100**	99	97	**100**	**100**
Place Dual Shoes	94	84	98	94	**100**	99	**100**	**100**	83	81	96	87	93	87
Place Empty Cup	99	95	98	99	99	97	98	96	98	99	94	**100**	**100**	**100**
Place Fan	75	75	93	87	94	89	96	84	84	92	**100**	**97**	97	93
Place Burger Fries	**100**	99	99	98	**100**	**100**	90	90	97	99	96	**100**	**100**	97
Place Mouse Pad	87	85	91	87	**99**	93	92	**94**	84	86	96	87	87	83
Place Object Basket	60	39	66	68	93	**96**	88	90	85	87	83	77	**97**	87
Place Object Scale	80	76	81	87	91	88	90	92	**93**	89	89	93	**93**	**97**
Place Object Stand	86	80	88	85	96	95	88	80	**97**	**96**	90	90	93	87
Place Phone Stand	91	85	98	97	99	96	**100**	**98**	75	80	90	97	97	97
Move Pillbottle Pad	81	81	87	86	**97**	97	82	72	96	**98**	**97**	90	**97**	97
Place Shoe	92	93	**99**	97	98	98	98	96	97	**99**	96	97	97	97
Press Stapler	87	83	93	98	85	82	96	96	94	90	90	**100**	**97**	**100**
Put Bottles Dustbin	84	79	81	79	87	91	72	70	85	**95**	**95**	83	87	83
Put Object Cabinet	80	79	88	71	85	87	74	74	66	76	**94**	**93**	93	90
Rotate Qrcode	89	87	89	73	**96**	**91**	90	84	84	83	93	90	87	90
Scan Object	72	65	67	66	96	91	60	64	86	79	89	**93**	**97**	87
Shake Bottle	99	97	**100**	97	**100**	97	**100**	98	99	99	**100**	**100**	**100**	**100**
Shake Bottle Horizontally	99	99	**100**	98	**100**	99	**100**	**100**	**100**	99	**100**	**100**	**100**	**100**
Stack Blocks Three	91	76	91	95	**99**	98	70	78	97	95	95	97	93	**100**
Stack Blocks Two	97	**100**	**100**	98	**100**	98	**100**	94	**100**	**100**	**100**	**100**	**100**	**100**
Stack Bowls Three	77	71	79	**87**	86	83	70	72	**88**	82	80	**87**	77	**87**
Stack Bowls Two	95	96	**98**	98	94	98	96	92	**98**	98	92	**100**	93	97
Stamp Seal	79	55	93	92	**96**	97	**96**	**98**	93	95	90	93	93	93
Turn Switch	62	54	**84**	78	44	45	82	**84**	61	72	61	60	80	70
Overall	82.74	76.76	88.66	87.02	92.90	91.50	86.36	85.04	89.76	90.68	91.88	90.32	**93.28**	**92.02**

Table 6. Task-level RoboTwin 2.0 results. Columns report clean/random success rates for each method.

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